

# XuanEr (Seona) Rao

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## RESEARCH INTERESTS

Human–Computer Interaction; Computational Design; Interactive Media; Gamification; Inclusive design; Digital Fabrication; AIGC

## EDUCATION

National University of Singapore, Singapore

Aug 2026–Jul 2027 (Expected)

Master of Science in Engineering Design & Innovation

Macau University of Science and Technology, Macau

Sep 2022–Jun 2026

Bachelor of Fine Arts in Digital Media (GPA: 3.94/4.00)

- **Graduated with First-Class Honors:** Awarded the highest honors with a Scholarship by Melco Resorts (Macau) limited to 2023, 2024, and 2025 respectively, Dean's Honors List (3 times)

## PUBLICATIONS

**Gamification and cognitive factors: Research hotspots, knowledge structure, and future directions based on bibliometric analysis. Journal of Intelligence**

Song, D., Guo, J., **Rao, XE.**, Hu, X., Gu, X., & Chen, J. (2026). *Journal of Intelligence*, 14(7), 150. [SSCI, JCR Q1] <https://doi.org/10.3390/jintelligence14070150>

## RESEARCH EXPERIENCE

Immersive Reality Lab, National University of Singapore

Research Assistant | Aug 2026–Present | Supervised by Dr Shaoyu Cai

- **VacGuide (leading; ongoing):** Investigating how passive internal guide geometry programs angular motion through vacuum-driven membrane collapse; developed 5-mm open-guide and multi-guide primitives spanning 0–180° rotations within a shared pneumatic pouch, and is characterizing geometry–motion relationships toward inverse design of articulated shape-changing structures.

- **Haptic Biofeedback Wearable For Therapistsupported Selfregulation (co-leading; ongoing):** Designing an EDA, PPG and IMU based wearable with vibrotactile and pneumatic feedback to support physiological self-regulation and therapist-mediated co-regulation during therapeutic conversations.

Digital Innovation and Humanities Education Research Center, Macau University of Science and Technology

Research Intern | Jun 2025–May 2026 | Supervised by Prof. Junming Chen

- Contributed to AIGC, Generative adversarial networks (GANs) for Bibliometric Review of AI-Generated Applications in Gamification Using CiteSpace. resulting work available as *Journal of Intelligence*, 14(7), 150. [SSCI, JCR Q1] <https://doi.org/10.3390/jintelligence14070150>

## WORK EXPERIENCE

NetDragon Websoft Inc. – Experience Designer (AIGC Production) | Jun-Sep 2024 Fujia, China

Qingbo Intelligent Technology Co.Ltd. – Design Intern (UIUX Design) | Apr-Jun 2024 Beijing, China

## SKILLS

- Prototyping & Fabrication: Pneumatic prototyping, 3D printing, heat sealing, soldering, electronics integration, rapid prototyping
- Hardware & Embedded: Arduino/ESP32, Lily Pad, E-textiles, sensors, circuit prototyping
- Programming & Design: Python, C/C++, LLM, MATLAB, Rhino, Unreal Engine 5, Unity, Figma, 3DMax, Blender

## Awards

- **2024 CMIT: National Gold Award**
- **16th User Experience Design Award** : First price of south china division (User experience Professional Association)
- **12th National College Digital Art&Design Awards:** Second price of Macau division
- **3rd Beijing Internet Audiovisual Arts Conference:** National third price